

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2017

EC210.01/EC210 Analog Communication

Date: 07.06.2017, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)** The output voltage of a transmitter is given by $400(1+0.4\sin 6280t) \sin 3.14 \times 10^7 t$ [07]
.This voltage is fed to a load of 600Ω resistance. Determine
- i. Carrier frequency
 - ii. Modulating frequency
 - iii. Carrier power
 - iv. Total power output
 - v. Peak power output

- Q - 2 (a)** Draw the circuit diagram of collector modulated class C amplifier? Explain its working in brief. [07]
- (b)** Derive the mathematical equation for current and voltage for amplitude modulation (AM). [07]

OR

- Q - 2 (a)** Derive the mathematical equation for average total power for sinusoidal amplitude modulation (AM). [07]
- (b)** How to detect amplitude modulated signals using diode detector? Explain it in brief. [07]
- Q - 3 (a)** How phase shift method improves the demerits of filter method? Explain in brief. [07]
- (b)** Draw and explain the block diagram of “Filter” method for single sideband suppressed carrier (SSBSC) generation. [07]

OR

- Q - 3 (a)** Draw and explain the circuit diagram of balance modulator using diode (ring modulator or lattice modulator) for DSBSC (double sideband suppressed carrier) generation. [07]
- (b)**
- i. What is vestigial sideband (VSB)? [02]
 - ii. Compare single sideband (SSB), double sideband (DSB), independent sideband (ISB) and vestigial sideband (VSB) with its technical aspects. [05]

SECTION – II

- Q - 4 (a)** Draw and explain frequency modulation circuit using varactor diode. [07]
- Q - 5 (a)** Draw and explain the block diagram of superheterodyne receiver with suitable signal spectra. [07]
- (b)** Explain frequency modulation with waveform. Also derive mathematical equation of frequency modulated (FM) wave. [07]

OR

- Q - 5 (a)** Explain gain and sensitivity of receiver in detail. [07]
- (b)** Explain foster seeley discriminator with necessary block diagram and phasor diagram. [07]
- Q - 6 (a)** What is an automatic gain control (AGC)? Explain different types of AGC control used in a communication receiver. [08]
- (b)** 1. The carrier swing of a frequency modulated signal is 70 KHz and the modulating signal is 7 KHz sine wave. Determine the modulation index of FM signal. [03]
2. In frequency modulating system the frequency deviation constant is 2 KHz/V. A sinusoidal modulating signal of amplitude 20V and frequency 4 KHz is applied. Calculate [03]
- i. Frequency deviation
- ii. Modulation index

OR

- Q - 6 (a)** Define all following terms. [08]
1. Frequency tuning range ratio
2. Image frequency
3. Capacitance tuning range ratio
4. Double spotting
- (b)** Why pre-emphasis and de-emphasis is used in FM transmitter and receiver? What is the function of pre-emphasis and de-emphasis in FM transmitter and receiver? [06]
